

Analysis Approaches for the Biomarker-by-Drug Interaction in Aggregated N-of-1 Trials: A Catalog

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Purpose. This paper lists every analysis approach for the biomarker-by-drug interaction that has been used, evaluated or proposed in this project. It covers:

- the published analysis of Hendrickson et al. (`orig`, commit 58b32a9) and its options;
- the nine specifications of manuscript 02;
- the test procedures of manuscripts 08 and 10, and the analysis strategies of manuscript 06;
- the summary-measures statistics of `docs/37`;
- the alternatives discussed in `docs/40`, `docs/41` and `docs/42`;
- the analyses in the covariance study (`08-covariance-study.R`), including its follow-up run.

The approaches are organized along the dimensions on which an analysis can vary (Section 2). The named approaches are then listed with their source, their status, and what is known about the size of their test (Sections 3 and 4). Section 5 summarizes which approaches are known to hold their size, and Section 6 lists the gaps.

Notation and glossary

Notation follows `analysis/report/NOTATION.md`.

- $\beta_{bm:D}$. The biomarker-by-drug interaction, the estimand.
- **Db, Dbc.** Binary drug state, D_{it} ; exposure-decayed drug indicator, $D_{bc,it}$ (1 on drug, $e^{-\lambda t_{sd}}$ off drug).
- **Size.** Type I error of the interaction test; nominal 0.05. An approach is *correctly sized* in a cell if its null rejection rate is within about 2 Monte Carlo standard errors of 0.05.
- κ . Mean model-based standard error over the empirical standard deviation of the estimate. Below 1 means anticonservative.
- **CS, AR(1).** Compound symmetry; first-order autoregressive correlation.
- **CR2.** Bias-reduced cluster-robust (sandwich) standard error, with Satterthwaite (Bell-McCaffrey) degrees of freedom.
- **KR.** Kenward-Roger standard-error and degrees-of-freedom correction.
- **MD.** Mancl-DeRouen bias-corrected sandwich for GEE.

- **cLDA.** Constrained longitudinal data analysis: baseline kept in the response with a common mean.
- **Construct.** The data-generating process a result was obtained under. *Published CS* is the 58b32a9 construct (**docs/36**); *configuration C* is separable AR(1) with graded coupling; *package* is the current package construct used by manuscripts 02, 06, 08 and 10.

1. Summary

1. **An analysis is a combination of about a dozen choices.** They are: the estimator family, the exposure coding, the time adjustment, the expectancy term, the baseline handling, the random effects, the working covariance, the standard error and degrees of freedom, the path term, the biomarker coding, the target coefficient, and the design-specific branches (Section 2).
2. **About 60 distinct approaches** have been used, evaluated or proposed (Section 3). Most vary one or two choices against a reference.
3. **Correct size depends mainly on the working covariance and the small-sample correction**, not on the exposure coding, the time adjustment or the baseline coding. Each of those moves the Type I error by at most about 0.02. The covariance and its correction move it by up to 0.10 (Section 4).
4. **Known to hold their size in every construct tested:**
 - an unstructured covariance with Kenward-Roger, baseline as a row (covariance study: 0.046 to 0.072);
 - CR2 standard errors with a random intercept and **corCAR1** (papers 02 and 10: 0.045 to 0.049);
 - GEE with the Mancl-DeRouen correction (papers 08 and 10).

Known to fail somewhere:

- the published random intercept under AR(1) data (Type I error 0.14 to 0.16 in CO);
 - any unstructured covariance with Satterthwaite only (0.072 to 0.104);
 - naive GEE (1.5 to 1.9 times nominal);
 - flexible time adjustments in Hybrid under carryover (about 0.07, under every covariance tested).
5. **Open:** the Kenward-Roger and CR2 versions of the baseline-covariate coding, CR2 with structured covariances under AR(1) data, and the mechanism behind the row coding's power under AR(1). All are in the follow-up run now in progress (Section 6).

2. Dimensions of an analysis

Each dimension lists its options, where each was used or proposed, and its current status. “Evaluated” means a simulation result exists; the source column gives the paper.

2.1 Estimator family

Option	Source	Status
Linear mixed model, conditional (<code>lmer</code> , <code>nlme::lme</code>)	orig, 02, 06, 08, 10, covariance study	evaluated
Marginal model with structured covariance (MMRM, <code>mmrm</code> , <code>glis</code>)	covariance study; docs/41, docs/42	evaluated
Summary measures: per-participant on-minus-off contrast, regressed on the biomarker (E9)	docs/37; 02 (G5); 08 (continuous-biomarker contrast)	evaluated
Strict repeated-measures ANOVA, dichotomized biomarker	docs/26; 08	evaluated (08)
GEE with sandwich variance	08; 10	evaluated
Two-stage approaches other than E9	08 (literature review)	not evaluated
Nonlinear component-matched model (Gompertz BR, PB, TV)	docs/31; 06	partly evaluated (06, recovery study)
Bayesian hierarchical N-of-1 model	02 (named as out of scope)	not evaluated
Randomization or permutation test	08 (literature)	not evaluated
Latent-class mixture model	manuscript 03	evaluated (different estimand)

2.2 Drug-exposure coding

Option	Code	Source	Status
Binary on-drug indicator Db	E1	orig; 02 G1	evaluated
Exposure-decayed Dbc, assumed half-life	E2	02 G3; implementations/original	evaluated

Option	Code	Source	Status
Binary plus lagged just-off-drug term L_{it}	E3	02 G2 (Jones-Kenward)	evaluated: no gain over E1
Binary plus linear time-since-discontinuation	E4	implementations/original (simplecarryover)	retired in 02
Lag term crossed with biomarker (L , $bm:L$)	E5	02, earlier S7	retired: worse than E1
t_{sd} crossed with biomarker	E6	02, earlier S7	retired: worse than E1
Dbc with half-life chosen by AIC	E7	02 G4	evaluated (one cell)
Dbc with alternative decay forms (Weibull, linear, power)		02 sensitivity blocks; NOTATION	evaluated as misspecification

2.3 Time adjustment

Option	Source	Status
None	E9 (docs/37)	evaluated
Linear $\mathbf{t}$	orig; 02; all mixed models above	evaluated
Linear $\mathbf{t}$ plus expectancy De	orig option (useDE); docs/40	population-mean limit only
Natural spline in $\mathbf{t}$, 3 df	docs/40, docs/42; covariance study	evaluated
Visit as factor	docs/40, docs/42; covariance study	evaluated
Phase indicators, phase by drug (attribution)	06 (phase-augmented)	evaluated (06): no bias reduction, less power
Random slope on $\mathbf{t}$	orig option ($\mathbf{t_random_slope}$)	not evaluated
Path-specific intercepts (E9 stage 2)	docs/37 (stratified E9)	closed form only

2.4 Expectancy

Option	Source	Status
Not modeled	orig as published	evaluated

Option	Source	Status
Design expectancy $\mathbf{D_e}$ as a fixed effect	orig option	population-mean limit only (docs/40)
Absorbed by visit effects	docs/40 , docs/42	evaluated
Measured belief covariate η	06 (balanced-placebo design)	evaluated (06)

2.5 Baseline handling

Option	Source	Status
Ninth response row, coded off drug	orig	evaluated
Excluded, no covariate	02 (all nine)	evaluated (02)
Covariate (ANCOVA)	docs/41 ; covariance study	evaluated with cs and us (Satterthwaite)
cLDA: row with common baseline mean, unstructured covariance	docs/41	equivalent to row plus us with visit effects
Change from baseline	docs/41	identical to ANCOVA in the construct
Baseline-by-drug moderator	docs/41	not evaluated (zero in the construct)
Row, with a separate biomarker effect at baseline	follow-up run	running

2.6 Random effects

Option	Source	Status
None (marginal covariance only)	MMRM; 10 (M0, M2)	evaluated
Random intercept	orig; 02; 10 (M1, M3)	evaluated
Random intercept and slope on $\mathbf{t}$	orig option	not evaluated
Random intercept, post-baseline shift and expectancy coefficient, phase-specific residual variance	this discussion (implied by the published construct)	not evaluated

2.7 Working covariance of the residuals

Option	Source	Status
Independent	orig; 10 (M1)	evaluated
corCAR1 (continuous-time AR(1))	02; 06; 08; 10 (M2, M3)	evaluated
Compound symmetry cs	covariance study (equals orig's random intercept)	evaluated
Heterogeneous CS csH	covariance study	evaluated
Unstructured us	covariance study; 10	evaluated
Spatial exponential sp_exp	covariance study	evaluated
AR(1), Toeplitz, ante-dependence	mmrm options	not evaluated (assume equal spacing)
Phase-specific variance (varIdent)	this discussion	not evaluated
GEE working correlation: independence, exchangeable, AR(1)	08; 10 (AR(1))	evaluated

2.8 Standard error and degrees of freedom

Option	Source	Status
Model-based SE, Satterthwaite df (lmerTest)	orig	evaluated
Model-based SE, containment df (nlme)	02; 06; 08	evaluated
Model-based SE, Satterthwaite df (mmrm)	covariance study	evaluated
Kenward-Roger	covariance study (us); 10	evaluated for us ; follow-up adds cs , csH and baseline covariate
CR2 sandwich, Satterthwaite df (clubSandwich)	02 (G6 to G9); 10	evaluated
CR2 sandwich, Satterthwaite df (mmrm)	follow-up run	running
CR0, CR1, CR3 sandwiches	10 (discussed)	not evaluated
Naive GEE sandwich	08; 10	evaluated
Mancl-DeRouen GEE sandwich	08; 10	evaluated

Option	Source	Status
OLS t , $N - 2$ df	E9	evaluated

2.9 Path or sequence term

Option	Source	Status
None	orig and most	evaluated
Fixed path effect	docs/37, single-path study	evaluated: no effect on <code>lmer</code> power
Path-specific intercepts in E9	docs/37	closed form only

2.10 Biomarker coding and target

Option	Source	Status
Uncentered biomarker	orig; 02	evaluated
Centered biomarker	docs/40; covariance study	evaluated (target unchanged)
Dichotomized biomarker	08 (strict RM-ANOVA)	evaluated: costs 0.08 to 0.21 power
Target <code>bm:Db</code>	orig; 02 G1, G2	evaluated
Target <code>bm:Dbc</code>	02 G3, G4	evaluated
Component-specific slopes	06	evaluated
Joint 2-df test (E5, E6)	02	retired

2.11 Design-specific branches

Option	Source	Status
Biomarker-by-time test when no participant varies in drug state (OL design): $Sx \sim bm + t + bm*t$	orig	evaluated in the published results; confounded with natural course and expectancy

3. Named approaches

Each row is a complete specification that has been fitted or proposed. Status is *evaluated*, *running* or *proposed*; the evidence column gives the construct and the key size result where one exists.

3.1 The published analysis and its options (orig, 58b32a9)

ID	Specification	Status	Evidence on size
O1	<code>lmer(Sx ~ bm + Db + t + bm*Db + (1 ptID))</code> , baseline row, Satterthwaite (published default)	evaluated	published CS: 0.049 to 0.058 (Hybrid), 0.04 to 0.05 (CO). Published results: 0.01 to 0.09. AR(1) data: 0.14 to 0.23 in CO (docs/36)
O2	O1 plus <code>De (useDE = TRUE)</code>	proposed	dropped by Hendrickson et al. for collinearity; correlation with <code>Db</code> 0.59 (docs/40)
O3	O1 with <code>(1 + t ptID)</code> (<code>t_random_slope = TRUE</code>)	proposed	
O4	OL branch: <code>Sx ~ bm + t + bm*t + (1 ptID)</code>	evaluated	confounded test; not a drug-by-biomarker interaction
O5	Later vendored implementation: <code>nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc)</code> , random intercept, <code>corCAR1</code>	evaluated as 02 G3	see 02 G3

3.2 Manuscript 02 (carryover specifications; package construct)

All use `nlme::lme` with a random intercept and `corCAR1`, no baseline row, linear `t`.

ID	Specification	Status	Evidence on size
G1	<code>Db</code> , target <code>bm:Db</code>	evaluated	mildly conservative (mean null 0.036 to 0.039)
G2	G1 plus lag <code>L</code>	evaluated as G1	no power gain over G1
G3	<code>Dbc</code> , target <code>bm:Dbc</code>	evaluated as G1	

ID	Specification	Status	Evidence on size
G4	G3, half-life chosen by AIC	evaluated	nominal despite selection (one cell)
G5	Paired difference (E9)	evaluated	
G6	G1, G2, G3, G4 with CR2	evaluated	CR2 restores nominal size (0.045 to 0.049 in 10)
G9			

3.3 Manuscripts 08 and 10 (test procedures and calibration; package construct)

ID	Specification	Status	Evidence on size
T1	Strict RM-ANOVA, dichotomized biomarker	evaluated	dichotomization costs 0.08 (08) to 0.21 power
T2	Continuous-biomarker within-participant contrast	evaluated	calibrated (08)
T3	<code>lme</code> , random intercept, <code>corCAR1</code> (production)	evaluated	conservative, 0.029 to 0.033 (08, 10 (10) M3)
T4	GEE, naive sandwich	evaluated	anticonservative, 1.5 to 1.9 (08, times nominal (08); 0.067 to 10) 0.069 (10)
T5	GEE, Mancl-DeRouen	evaluated	restores nominal size (08, 10)
M0	OLS, no within-participant correlation	evaluated	not anticonservative (10)
M1	Random intercept only	evaluated	well calibrated in 10's (10) construct
M2	<code>corCAR1</code> only	evaluated	intermediate (10)
M3	Random intercept plus <code>corCAR1</code>	evaluated	conservative; CR2 fixes it; (10) KR inert on it
M-us	Unstructured, KR against Satterthwaite	evaluated	Satterthwaite 0.076, KR (10) 0.059

3.4 Manuscript 06 (component decomposition)

ID	Specification	Status	Evidence
C1	One-component (bm:Dbc)	evaluated	unbiased if the biomarker is unrelated to PB and TV; size 0.031 to 0.056
C2	Phase-augmented: phase, phase by drug, phase by biomarker by drug	evaluated	no bias reduction; less power; size 0.038 to 0.049
C3	Belief-covariate decomposition	evaluated	recovers the pharmacological slope only on a balanced-placebo design
C4	Blinded-stratum contrast; Gompertz-basis decomposition	evaluated	fail on the Hybrid design

3.5 Summary measures (docs/37)

ID	Specification	Status	Evidence
S1	E9: post-baseline on-minus-off contrast, OLS on biomarker	evaluated (closed form and Monte Carlo)	exact closed form; validated over 566 cell-variants
S2	E9b: baseline counted as off drug	evaluated	mirrors O1's baseline coding; 15% to 16% noisier than E9 without carryover (docs/41)
S3	Stratified E9: path-specific intercepts	closed form only	removes the CO period-effect loss

3.6 This discussion (docs/40 to docs/42; published CS construct)

ID	Specification	Status	Evidence on size
D1	O1 with ns(t, 3)	evaluated	Hybrid $t^{1/2}$ 1: 0.073 (O1 0.058)
D2	O1 with visit factor	evaluated	Hybrid $t^{1/2}$ 1: 0.080
D3	O1 plus fixed path effect	evaluated	power changes by at most 0.001
D4	O1 on a single path against the mixture	evaluated	no material difference in Hybrid

ID	Specification	Status	Evidence on size
D5	Randomized (multinomial or permuted-block) path allocation	proposed	
D6	Random intercept, post-baseline shift and expectancy coefficient, varIdent by phase	proposed	
D7	Baseline-by-drug moderator	proposed	

3.7 The covariance study: main run (published CS and configuration C)

All **mmrm**, centered biomarker, binary **Db**. Fitted to the same trials: 16 cells, 250 alternative and 500 null replicates.

ID	Coding	Covariance	Time	Test	Size range
V1	row	cs	linear, spline, visit	Satterthwaite	CS 0.036 to 0.072; configuration C, CO 0.140 to 0.158
V2	row	csH	linear, spline, visit	Satterthwaite	as V1 (CO, configuration C: 0.138 to 0.164)
V3	row	us	linear, spline, visit	Satterthwaite	0.072 to 0.104
V4	row	sp_exp	linear, spline, visit	Satterthwaite	0.026 to 0.062; low power under CS
V5	row	us	linear, visit	Kenward-Roger	0.046 to 0.072
V6	covariate	cs	linear, spline, visit	Satterthwaite	CS 0.034 to 0.064; configuration C, CO 0.146 to 0.164
V7	covariate	us	linear, spline, visit	Satterthwaite	0.072 to 0.096

3.8 The covariance study: follow-up run (running; same trials)

ID	Coding	Covariance	Time	Test	Question
F1	covariate	us	linear, spline, visit	Kenward-Roger	is the covariate coding correctly sized?
F2	covariate	cs	linear, visit	Kenward-Roger	does KR change a parsimonious model?
F3	row	us	spline	Kenward-Roger	completes V5
F4	row	cs, csh	linear, visit	Kenward-Roger	can KR repair a misspecified covariance?
F5	row	cs, csh, us, sp_exp	linear, visit	CR2	does CR2 hold size under every covariance?
F6	covariate	cs, us	linear, visit	CR2	as F5, covariate coding
F7	row, separate baseline biomarker effect	us	visit	Kenward-Roger	does the shared biomarker effect explain the row coding's power?

4. What is known about size

Main conclusions:

- 1. The covariance and its small-sample correction dominate.** Changing the exposure coding (02), the time adjustment (docs/40) or the baseline coding (covariance study) moves the Type I error by at most about 0.02. Changing the working covariance or the correction moves it by up to 0.10.
- 2. A parsimonious covariance that is wrong fails in either direction.** The published random intercept is anticonservative under AR(1) data in CO (0.14 to 0.16). The random intercept with `corCAR1` is conservative in the package construct (0.029 to 0.039). CR2 restores the latter. Kenward-Roger does not, being inert on parsimonious structures (10).
- 3. An unstructured covariance needs Kenward-Roger.** With Satterthwaite alone it is anticonservative in every construct tested (0.072 to 0.104 in the covariance study, 0.076 in 10). Kenward-Roger brings it to nominal.

4. **Sandwich estimators need small-sample corrections.** Naive GEE is anticonservative. Mancl-DeRouen and CR2 restore the size.
5. **A residual size problem remains for flexible time adjustments in Hybrid under carryover** (about 0.07). It persists under every covariance tested, including `us` with Kenward-Roger, and its cause is unexplained.

Construct dependence. Results come from three constructs: published CS, configuration C, and the package construct. Approaches have not all been tested in all three. The parsimonious covariances are the most construct-dependent: the random intercept is well calibrated under CS data (10 M1; covariance study V1 under CS) and fails under AR(1) data.

5. Approaches correctly sized in every cell tested

Approach	Constructs	Size range	Power relative to the published analysis
Row, <code>us</code> , Kenward-Roger, linear or visit (V5)	published CS, configuration C	0.046 to 0.072	similar under CS; +0.10 to +0.12 in Hybrid under configuration C
Random intercept plus <code>corCAR1</code> , CR2 (02 G6 to G9; 10)	package	0.045 to 0.049	slightly higher than model-based (recovers its conservatism)
GEE, Mancl-DeRouen (T5)	package	nominal	modest power cost against the mixed model (08)
<code>sp_exp</code> , Satterthwaite (V4)	published CS, configuration C	0.026 to 0.062	much lower under CS (0.31 to 0.52 against about 0.70)

V5's upper value of 0.072 occurs with the visit-factor time adjustment in Hybrid at $t^{1/2} = 1$ (published CS), the residual problem of Section 4, point 5. With linear time V5 stays within 0.052 to 0.066.

6. Gaps and the evaluation in progress

Being answered by the follow-up run (F1 to F7): Kenward-Roger and CR2 for the baseline-covariate coding; whether CR2 rescues the misspecified parsimonious covariances under AR(1) data; whether Kenward-Roger does (expected not); and the mechanism behind the row coding's power under AR(1).

Not yet scheduled:

1. The cause of the flexible-time size excess in Hybrid under carryover (Section 4, point 5).
2. The random-effects structure implied by the published construct, with phase-specific variance (D6).
3. A random slope on time (O3) and the expectancy term (O2) by simulation, not only in the population-mean limit.
4. GEE with Mancl-DeRouen and the summary-measures statistic in the published CS and configuration C constructs, for a direct comparison with V5.
5. Exposure-weighted coding (Dbc) crossed with the covariance choices. Manuscript 02 evaluated it only under corCAR1.
6. Randomized path allocation (D5).
7. Power at matched size, for every approach found correctly sized.
8. Bayesian hierarchical models and randomization tests, named in the literature reviews of 02 and 08 and not evaluated.

Suggested core comparison. Once the follow-up run is in, a single paired study in the three constructs could cover:

- the approaches correctly sized so far (V5, CR2 variants, GEE with Mancl-DeRouen);
- the published analysis as reference (O1);
- crossed with the exposure coding (Db against Dbc) and the baseline coding.

That would give one table of size and power from which a pre-specified primary analysis for a Hybrid trial could be chosen.

7. Sources

- orig: `github.com/rchendrickson/pmsimstats`, commit 58b32a9, `R/lme_analysis.R` and vignette 1 (inspected).
- Manuscripts: `analysis/report/02-carryover-sensitivity/`, `06-component-decomposition/`, `08-test-procedure-design-sensitivity/`, `10-interaction-test-calibration/` (read for their specifications and headline results; the numbers quoted are those reported in the manuscripts).
- Project documents: `docs/26`, `docs/31`, `docs/36`, `docs/37`, `docs/40`, `docs/41`, `docs/42`.
- Covariance study: `analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R`; results in `analysis/data/quick-sim/carryover-closed-form/covariance-study/summary-alt250-n` (verified); follow-up running.